

MATH 115A: LINEAR ALGEBRA
2024 SPRING - LECTURE 6
Jukka Keränen

MIDTERM 2 PRACTICE EXAM

Your Name

The Last **Six** Digits of Your Student ID number

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By signing below, you confirm that you did not cheat on this exam.

INSTRUCTIONS

- Please do **not** open this booklet until you are told to do so.
- Your cell phone must be **turned off completely** and stowed away where you cannot access it.
- You are not allowed to access the Internet during the exam.
- No books or notes are allowed, except for one 3" x 5" index card.
- If you have a question at any time during the exam, please raise your hand.
- Make sure to show all your work and justify your answers fully.
- Make sure to write legibly. **Illegible work will not be graded.**
- If you are using a pencil, make sure your writing is dark and easily visible.
- Any work you cross out will not be graded.

FRIENDS, GOOD LUCK! YOU'VE GOT THIS!

1. (a) State the definition of a *linear map*.

A linear map $f: V \rightarrow W$ would be one in which

$$(1) \forall x, y \in V \quad f(x) + f(y) = f(x+y)$$

$$(2) \forall a \in F \ \& \ \forall x \in V \quad f(ax) = a f(x)$$

(b) Let $f: V \rightarrow W$ be a linear map. Show that $f(\mathbf{0}_V) = \mathbf{0}_W$.

$$f(\mathbf{0}_V) = f(\mathbf{0}_V)$$

$$f(\mathbf{0}_V \cdot 1) = f(\mathbf{0}_V)$$

$$f(\mathbf{0}_V) \cdot \mathbf{0}_W = f(\mathbf{0}_V)$$

$$\mathbf{0}_W = f(\mathbf{0}_V)$$

$$f(\mathbf{0}_V) = -f(\mathbf{0}_V)$$

$$f(\mathbf{0}_V + \mathbf{0}_V) = f(\mathbf{0}_V)$$

$$f(\mathbf{0}_V) + f(\mathbf{0}_V) = f(\mathbf{0}_V)$$

$$f(\mathbf{0}_V) + (-f(\mathbf{0}_V)) + f(\mathbf{0}_V) = f(\mathbf{0}_V) + (-f(\mathbf{0}_V))$$

$$\mathbf{0}_W + f(\mathbf{0}_V) = \mathbf{0}_W$$

$$f(\mathbf{0}_V) = \mathbf{0}_W$$

2. (a) Let V and W be finite-dimensional vector spaces, and let $f : V \rightarrow W$ be a linear map. State the *Rank-Nullity theorem*.

$$\dim_F(\text{im}(f)) + \dim_F(\ker(f)) = \dim_F(V)$$

(b) Show that if f is surjective, then $\dim(V) \geq \dim(W)$.

Suppose f is onto, i.e. $\forall y \in W$

$$\exists \text{ a } x \in V \text{ s.t. } f(x) = y$$

$$\dim_F(\text{im}(f)) + \dim_F(\ker(f)) = \dim_F(V)$$

$$\dim_F(W) + \dim_F(\ker(f)) = \dim_F(V)$$

$$\underline{\dim(\ker(f))} = \dim_F(V) - \dim(W) \geq 0$$

$$\dim_F(V) \geq \dim(W)$$

(c) Suppose that $V = \text{span}(\{v_1, \dots, v_m\})$. Show that $\text{im}(f) = \text{span}(\{f(v_1), \dots, f(v_m)\})$.

let $f(x) \in \text{im}(f)$ with $x \in V$

(C)

$$\bar{x} = a_1 v_1 + \dots + a_m v_m$$

$$f(\bar{x}) = f(a_1 v_1 + \dots + a_m v_m)$$

$$f(\bar{x}) = a_1 f(v_1) + \dots + a_m f(v_m)$$

$$f(x) \in \text{span}(\{f(v_1), \dots, f(v_m)\})$$

(=) $w \in \text{span}(\{f(v_1), \dots, f(v_m)\})$

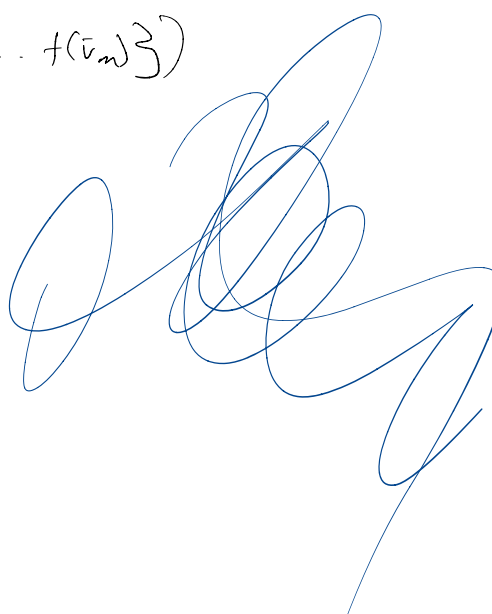
$$\text{Then } \exists a_1, \dots, a_n \in F \text{ s.t.}$$

$$w = a_1 f(v_1) + \dots$$

$$= f(a_1 v_1 + \dots)$$

$$\text{since } v_1, \dots, v_m \in V,$$

$$a_1 v_1$$



Alternatively:

2. State and prove the Rank-Nullity theorem.

3. Let $f : P_2(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ be a linear map given by

$$f(c + bx + ax^2) = \begin{pmatrix} a+b & b-c \\ 2c & -b \end{pmatrix}$$

Consider the basis $\mathfrak{B} = \{1, x, x^2\}$ of $P_2(\mathbb{R})$ and the basis $\mathfrak{C} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ of $M_{2 \times 2}(\mathbb{R})$.

(a) Find $[f]_{\mathfrak{B}}^{\mathfrak{C}}$.

$$[f]_{\mathfrak{B}}^{\mathfrak{C}} = \begin{pmatrix} [f(1)]_{\mathfrak{C}} & [f(x)]_{\mathfrak{C}} & [f(x^2)]_{\mathfrak{C}} \end{pmatrix}$$

$$f(1) = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix} \Rightarrow [f(1)]_{\mathfrak{C}} = \begin{pmatrix} 0 \\ -1 \\ 2 \\ 0 \end{pmatrix}$$

$$f(x) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 1 \\ 0 \\ -1 \end{pmatrix}$$

$$f(x^2) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$[f]_{\mathfrak{B}}^{\mathfrak{C}} = \begin{pmatrix} 0 & 1 & 1 \\ -1 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

(b) Give two independent reasons why f is **not** a bijection.

f isn't one-to-one as $\ker(f) \neq \{0\}$

f isn't onto as $\dim(\text{Im}(f)) < 4$

$$\dim_{\mathbb{R}}(\text{Im}(f)) \leq 3$$

4. Consider the linear map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$f((a, b)) = (a + b, a - b).$$

Consider the bases $\mathfrak{B} = \{(1, 2), (2, 1)\}$ and $\mathfrak{S} = \{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$.

(a) Find the change of basis matrix from $\mathfrak{B}$ to $\mathfrak{S}$.

$$\begin{aligned} [I_V]_{\mathfrak{S}}^{\mathfrak{B}} &= \begin{pmatrix} (1, 2)_{\mathfrak{S}} & (2, 1)_{\mathfrak{S}} \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \end{aligned}$$

(b) Find $[f]_{\mathfrak{S}}^{\mathfrak{S}}$.

$$\begin{aligned} [f]_{\mathfrak{S}}^{\mathfrak{S}} &= \begin{bmatrix} f(1, 0)_{\mathfrak{S}} & f(0, 1)_{\mathfrak{S}} \end{bmatrix} \\ &= \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{aligned}$$

(c) Find $[f]_{\mathfrak{B}}^{\mathfrak{B}}$ in terms of your answers to parts (a) and (b). You do not need to carry out the matrix multiplication.

$$[f]_{\mathfrak{B}}^{\mathfrak{B}} = [I_V^{-1}]_{\mathfrak{S}}^{\mathfrak{B}} \cdot [f]_{\mathfrak{S}}^{\mathfrak{S}} \cdot [I_V]_{\mathfrak{B}}^{\mathfrak{S}}$$